SALMAN AKHATAR

Senior Data Scientist: AI/ML Engineer

Seasoned Data Scientist, having experience with Full Stack Software Engineer stacked with staggering 9+ years of professional experience in developing scalable, high-performance projects. Skilled in Python, FastAPI, Django, GenAi, LLM, ML, and JavaScript, with expertise spanning machine learning, backend and frontend development. Known for delivering innovative, impact- driven projects that enhance business efficiency and drive technological growth. I am dedicated to leveraging my skills in a dynamic, growth- oriented environment, contributing to cutting-edge software solutions alongside forward-thinking professionals.

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PROFESSIONAL SUMMARY

- Senior Full Stack Developer with 9+ years of experience in machine learning, backend and frontend development, specializing in Python, JavaScript, andframeworks like Django, Flask, and FastAPI.
- Backend Expertise: Proficient in database design, data modeling, and API development, with a focus on creating efficient, scalable solutions.
- LLM and Generative AI Proficiency: Skilled in integrating Large Language Models and Generative AI technologies to drive intelligent, automated solutions.
- Leadership in Project Delivery: Demonstrated success in managing cross-functional teams, leading and mentoring junior engineers, driving projects to completion, and enhancing client relationships.
- Award-Winning Performance: Recognized for excellence with awards such as "Collaborative Extraordinaire" and "I Am Exceptional."
- Advanced DevOps Skills: Skilled in deploying solutions using Docker and Kubernetes, and in managing CI/CD pipelines on cloud platforms like AWS and Azure.

SKILLS

Core Competencies: Python, FastAPI, Django, MySQL, PostgreSQL, ORM, Flask, LLM, RAG, Vector DB, GenAI, Machine Learning

Soft Skills: Communication, Leadership, Teamwork, Problem-solving

TECHNICAL SKILLS

Microservices Architecture | RESTful APIs | Docker | Python | Git | FastAPI | LLM | Django | Kubernetes | JavaScript | Machine Learning | Flask | GenAi | Prompt Engineering | NLP | Web Scraping | PostgreSQL | Elasticsearch | Microservices | Docker | Podman | AWS / Azure / GCP | Pyspark | Pandas | Kafka | Airflow | Celery | CI/CD | ORMs (Django, SQL Alchemy)

WORK EXPERIENCE

Senior Data Scientist: AI/ML Engineer

IBM *(Feb 2025 – Present)*

Working on different LLM related projects.

- Working on FastAPI

Senior Consultant

EY LLP *(Nov 2021 – Feb 2025)*

Led the development and deployment of SAM and cybersecurity solutions, ensuring effective team collaboration and project delivery. Architected and executed end-to-end software development projects, effectively managing cross-functional teams to meet project milestones.

- Received Collaborative Extraordinaire recognition for exceptional performance in Q2 2024.
- Promoted to Senior Consultant in Q4 2023, following a series of successful project completions and leadership initiatives.
- Oversaw the full development cycle for three key modules in a major cybersecurity project.
- Participated in EY's New Senior Program focused on leadership and inclusivity training.
- Implemented RBAC and token-based authentication for secure API middleware access.
- Managed Docker-based deployments on Docker Hub and virtual machines, ensuring seamless distribution.
- Supervised CI/CD pipelines for deployments on Azure VMs, enhancing deployment efficiency.
- Developed data transformation processes to handle millions of records in PostgreSQL using FastAPI.
- Built and deployed two chatbot wrappers utilizing Large Language Models (LLM) to support client automation.
- Received the 'I Am Exceptional' award for outstanding performance in Q1 2023.
- Developed BOT software for automated data extraction across multiple operating systems, deployed on over 20,000 machines.
- Boosted data processing speeds by integrating Redis, leading to optimized performance.
- Built robust Python-based APIs for seamless project integration.
- Designed high-performance ETL processes capable of handling millions of records efficiently.
- Directed deployments across development, QA, and UAT environments, ensuring high-quality standards and stability.

Achievements:

- Awarded Collaborative Extraordinaire recognition in Q2 2024.
- Promoted to Senior Consultant in Q4 2023.
- Awarded 'I Am Exceptional' recognition for Q1 2023 performance.

Software Engineer III

TELUS International *(Dec 2018 - Nov 2021)*

Developed and delivered Python-based applications and automated data workflows to streamline operations.

- Engineered five high-impact Python projects using Flask and Django frameworks.
- Built multiple ETL processes with data preprocessing for BigQuery, improving data accessibility.
- Automated ETL workflows through Airflow, enhancing data pipeline efficiency.
- Collaborated closely with architects and analysts to drive end-to-end project development.
- Established best practices for quality and cost-effective delivery within the organization.
- Strategized product roadmaps in consultation with senior leadership to align with future goals.

Achievements:
- Awarded 'Pat on the Back' for outstanding performance in Q2 2019.

Senior Software Engineer
App & Web Solutions Pvt Ltd *(Jan 2018 - Dec 2018)*

Led software development and integration initiatives, optimizing internal processes to ensure quality deliverables.

- Delivered five projects and sub-modules, including a CNN-based image processing application.
- Developed APIs in Python and PHP to support both UI and mobile applications.
- Streamlined workflows by implementing standardized technical procedures.
- Troubleshot and resolved technical issues, enhancing customer satisfaction.
- Developed and supported UI schema objects, reports, and dashboards for enhanced data visualization.
- Documented and tested software throughout the development lifecycle to maintain quality.

Senior Software Engineer
Roundone Network Pvt Ltd *(May 2017 - Jan 2018)*

Designed and led development for two key projects and several sub-modules, including CRM panels, achieving an 85% increase in productivity and reporting efficiency.

- Promoted from Software Engineer to Senior Software Engineer within two months due to exceptional performance.
- Developed applications, CRMs, and APIs using Python and PHP, delivering scalable and maintainable solutions.
- Created and optimized data crawlers to fetch job listings, performing multi-level data preprocessing for platform use.
- Planned and implemented models, interfaces, and workflows to enhance scalability and code extensibility.
- Collaborated with product management and architecture teams to align features with business requirements.
- Led a team of 5-7 engineers, providing technical direction while managing personal tasks and responsibilities.
- Conducted technical discussions, shared ideas via whiteboarding, and participated in technical design reviews.

Achievements:
- Promoted to Senior Software Engineer within two months of joining due to exemplary contributions.

Software Engineer
Piana IT Solutions Pvt Ltd *(Jan 2016 - May 2017)*

Contributed to full-stack development for over four projects plus sub-modules, meeting stakeholder requirements.

- Partnered with CEOs and stakeholders to define technical specifications aligned with business needs.
- Authored high-quality code, leading testing and debugging efforts to ensure reliable software performance.
- Adhered to the full software development lifecycle process, maintaining best practices.
- Pursued continuous learning, exploring new technologies to improve software delivery.
- Evaluated solutions and optimized project performance for enhanced efficiency.

EDUCATION

M. Sc. Tech Industrial Maths with Computer Application (Jul 2013 - Jun 2016)
Jamia Millia Islamia, New Delhi, India

B. Sc. Physical Science with Computer Science (Jul 2010 - Jun 2013)
University of Delhi, New Delhi, India

ACHIEVEMENTS

- Received 'Collaborative Extraordinaire' recognition for outstanding teamwork and excellence in project delivery.
- Successfully led the development and deployment of EY Atom Bot, resulting in deployment on over 20,000 machines, showcasing significant scalability and reliability.

NOTABLE PROJECTS

IBM: Super Agents
- LLM enabled super agents, which talk to multiple single purpose agents and creates a dynamic multi-agent flow to solve end-to-end problems.

EY Atom Bot & Atom Bot Server: Enterprise System Monitor
- **Atom Bot:** Developed and deployed an executable automation bot across 25,000+ machines, streamlining data processing through optimized API call handling and robust ETL pipelines. Implemented unit testing and thorough documentation to ensure high reliability and maintainability.
- **Atom Bot Server:** Engineered a scalable backend server capable of processing thousands of simultaneous requests from Atom Bot, facilitating efficient data management and seamless communication across systems.
- **Technologies:** Python3+, ETL Pipelines, API Design, PostgreSQL, Microservices, Automation, Docker / Podman, Azure, ReactJS
- **Outcome:** Streamlined data management across systems, ensuring high reliability and reducing manual data handling time by 40%.

EY Atom ETL Engine
- **Automated ETL Tool**: Developed a robust ETL engine to automate data ingestion and processing from 15+ connectors or data sources into a PostgreSQL database. Engineered complex calculations to handle data transformation requirements and ensure seamless integration with EY Atom Bot.
- **Technologies:** Python3+, FastAPI, ETL Pipelines, Data Processing, PostgreSQL, Automation, Docker, ReactJS
- **Outcome:** Processed millions of records daily with 99.9% accuracy, reduced data processing time by 70% and implemented automated data quality checks with 95% issue detection.

EY Quark: Cybersecurity Solution

- **Comprehensive Cybersecurity Platform:** Developed a scalable, microservices-based cybersecurity product incorporating agentless data discovery (like system information, installed products, software metering, services, process, keywords etc.), application rollout, patch management, and configuration management.
- **API-Driven Architecture:** Leveraged Flask and FastAPI to build distributed, API-based solutions for modular functionality and streamlined communication across components.
- **Automation and Compliance:** Integrated Ansible for process automation and implemented misconfiguration detection and remediation protocols to enhance CIS security compliance and reduce manual intervention.
- **Technologies:** Django, FastAPI, Ansible, Microservices Architecture, Docker
- **Outcome:** Enhanced organizational security with automated misconfiguration remediation and streamlined patch management, reducing risk exposure and increasing system uptime by 20%.

EY Quasar: LLM-Powered Data Query and Visualization Tool

- **Intelligent Data Query Engine:** Developed an LLM-based wrapper using Ollama and Mistral (7b parameters) to interpret natural language inputs and generate optimized SQL queries, enabling non-technical users to retrieve complex data insights.
- **Automated Visualization:** Designed the system to generate dynamic visual representations (e.g., bar, line, and donut charts) directly from SQL output, with each graph accompanied by auto-generated titles and concise summaries for improved data interpretation.
- **Technologies:** Python3+, PostgreSQL, LLM (Ollama, Mistral), Dfog, Data Visualization
- **Outcome:** Enabled non-technical users to access complex data insights, improving report generation time by 30% and enhancing decision-making with clear visual summaries.

EY Qbot: Document Intelligence System

- **LLM-Powered Document Processing:** Developed an advanced LLM wrapper to process and store document data in ChromaDB, enabling users to query document content directly with natural language.
- **Contextualized Responses with References:** Engineered the system to generate context-aware responses, providing specific page numbers and text chunks as references, enhancing traceability and accuracy.
- **Dynamic Summarization:** Designed Qbot to not only retrieve data from documents but to synthesize responses, delivering well-formed answers that facilitate comprehensive document understanding, saving a lot of time for the team and people which earlier required hours to retrieve same data.
- **Technologies:** Python3+, PostgreSQL, LLM (Ollama, Mistral), ChromaDB for vector storage
- **Outcome:** Successfully processed and indexed over 1000s of documents while achieving 95% accuracy in source attribution, resulting in reduced document query response time by 70%.

Telus: Relevant Resume Finder

- **NLP-Powered Resume Matching Tool:** Engineered an NLP-based solution to analyze and match job descriptions with a repository of over 20,000 resumes in various formats (PPT, DOCX, PDF), providing ranked results based on relevance scores.
- **Enhanced Recruitment Efficiency:** Designed to streamline candidate sourcing by returning resumes with relevance percentages, enabling recruiters to make data-driven hiring decisions faster and more effectively.
- **Technologies:** Python3+, Web Scraping, Django REST framework, NLP
- **Outcome:** Successfully processed 20,000+ resumes while achieving 90% accuracy in relevance matching, resulting in reduced manual resume screening time by 85%.

LANGUAGES

English, Arabic, Urdu, Hindi

HOBBIES

- **Building Personal Tech Projects**: Experimenting with new technologies and frameworks to develop innovative solutions.
- **Open-Source Contributions**: Actively contributing to community-driven projects to support the tech ecosystem.
- **Industry Engagement**: Attending technology conferences and meetups to stay updated on industry trends.
- **Reading**: Exploring books on technology, leadership, and personal development.
- **Socializing**: Enjoys connecting with friends and engaging in collaborative and creative discussions.